

**Problem 6**

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**Problem 1** Find the derivative of the following functions:

(a)  $f(x) = \sin(x) \cos(x)$

(b)  $f(x) = \frac{e^x \tan(x)}{\sec(x) + 2}$

(c)  $f(x) = e^{x \tan(x)}$

(d)  $f(x) = \sin(x) \cos(x) e^{3x}$

(e)  $f(x) = \frac{x + 5}{7x^6 + \cot(x)}$

(f)  $f(x) = \sin(2x) \sec^3(x^2 + 4x)$